Introduction to Machine Learning

A short practice document for testing a Retrieval-Augmented Generation (RAG) chatbot.

1. What is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computer systems to identify patterns in data and use those patterns to make predictions or decisions. Instead of writing every rule by hand, developers train a model using examples. The model learns relationships from the training data and applies what it has learned to new data.

2. Main Types of Machine Learning

Supervised learning uses labelled examples. Each training example contains input data and a known target. Classification predicts a category, such as whether an email is spam. Regression predicts a numerical value, such as the expected price of a house.

Unsupervised learning works with data that does not have target labels. The algorithm looks for structure or groups in the data. Clustering customers according to similar purchasing behaviour is one example.

Reinforcement learning involves an agent that learns by taking actions in an environment. The agent receives rewards or penalties and tries to learn a strategy that maximizes its total reward. It can be used in robotics and game-playing systems.

3. Training and Testing

A dataset is commonly divided into training and testing sets. The training set is used to fit the model. The testing set is kept separate and is used to estimate how well the trained model performs on data it has not seen before. A validation set may also be used during model development to compare settings and select a model.

4. Overfitting and Underfitting

Overfitting happens when a model learns the training examples too closely, including noise or accidental details. It may perform very well on training data but poorly on new data. Underfitting happens when a model is too simple to learn the important patterns, so its performance is poor on both training and testing data.

5. Common Evaluation Metrics

Accuracy is the proportion of predictions that are correct. Precision measures how many items predicted as positive are actually positive. Recall measures how many actual positive items the model successfully identifies. For regression, mean absolute error (MAE) calculates the average absolute difference between predicted and actual values.

6. Example: Email Spam Detection

A spam detection system can be trained using emails labelled as spam or not spam. The model may learn patterns involving words, links, sender information, and message structure. When a new email arrives, the model predicts a class. The system can then place likely spam messages in a separate folder. The

prediction is not guaranteed to be correct, so users should be able to review messages classified as spam.

Practice Questions

1. What is machine learning?
2. How does supervised learning differ from unsupervised learning?
3. What is one example of a regression task?
4. Why should testing data be kept separate from training data?
5. What is the difference between overfitting and underfitting?
6. Which metrics can be used to evaluate a classification model?